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```
lokeshtella@LAPTOP-0PA61QEN:~/OSSP$ mkdir src include obj bin docs tests scripts assets
lokeshtella@LAPTOP-0PA61QEN:~/OSSP$ touch README.md LICENSE Makefile .gitignore
lokeshtella@LAPTOP-0PA61QEN:~/OSSP$ touch src/main.c
touch src/shell.c
touch src/parser.c
touch src/executor.c
touch src/builtin.c
lokeshtella@LAPTOP-0PA61QEN:~/OSSP$ touch include/shell.h
touch include/parser.h
touch include/executor.h
touch include/builtin.h
lokeshtella@LAPTOP-0PA61QEN:~/OSSP$ touch docs/design.md
touch docs/report.pdf
lokeshtella@LAPTOP-0PA61QEN:~/OSSP$ touch tests/test_parser.c
touch tests/test_executor.c
lokeshtella@LAPTOP-0PA61QEN:~/OSSP$ touch scripts/run.sh
lokeshtella@LAPTOP-0PA61QEN:~/OSSP$ touch assets/architecture.png
lokeshtella@LAPTOP-0PA61QEN:~/OSSP$ sudo apt install tree
```

```
lokeshtella@LAPTOP-0PA61QEN:~$ cd ~
lokeshtella@LAPTOP-0PA61QEN:~$ pwd
/home/lokeshtella
lokeshtella@LAPTOP-0PA61QEN:~$ mkdir OSSP
lokeshtella@LAPTOP-0PA61QEN:~$ cd OSSP
lokeshtella@LAPTOP-0PA61QEN:~/OSSP$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/lokeshtella/OSSP/.git/
```

```
lokeshtella@LAPTOP-0PA61QEN:~/OSSP$ tree
```

```
.
├── LICENSE
├── Makefile
├── README.md
├── assets
│   └── architecture.png
├── bin
├── docs
│   ├── design.md
│   └── report.pdf
├── include
│   ├── builtin.h
│   ├── executor.h
│   ├── parser.h
│   └── shell.h
├── obj
├── scripts
│   └── run.sh
├── src
│   ├── builtin.c
│   ├── executor.c
│   ├── main.c
│   ├── parser.c
│   └── shell.c
└── tests
    ├── test_executor.c
    └── test_parser.c
```

```
9 directories, 18 files
```

```
lokeshtella@LAPTOP-0PA61QEN:~/OSSP$
```

1

2)

```
lokeshtella@LAPTOP-0PA61QEN:~/OSSP$ nano fork_exec.c
lokeshtella@LAPTOP-0PA61QEN:~/OSSP$ gcc fork_exec.c -o fork_exec
lokeshtella@LAPTOP-0PA61QEN:~/OSSP$ ./fork_exec
This is the Parent Process.
Parent PID = 631
Child PID = 632
This is the Child Process.
Child PID = 632
lokeshtella@LAPTOP-0PA61QEN:~/OSSP$
```

```
#include <stdio.h>
#include <unistd.h>
int main() {
    pid_t pid;
    pid = fork();    // Create a new process
    if (pid < 0) {
        printf("Fork failed!\n");
    }
    else if (pid == 0) {
        // Child Process
        printf("This is the Child Process.\n");
        printf("Child PID = %d\n", getpid());
    }
    else {
        // Parent Process
        printf("This is the Parent Process.\n");
        printf("Parent PID = %d\n", getpid());
        printf("Child PID = %d\n", pid);
    }
    return 0;
}
```